

Week 10 HW04 ???

?? ??? ?? ??? ?? ??: ???(CNN)? ???(RNN/CNN) ??

1. ?? ???

??? ??? MNIST

??? ?? ??? MNIST ??? ?? ????. ?? ??? 28 x 28 grayscale ????, ??? ?? 0?? 9?? ? 10???. ?? ??? 55,000?, ?? ??? 5,000?, ??? ??? 10,000??? ???.

??? ??? NSMC

??? ?? ?? ??? ?? ?? ?? ?? ???(NSMC)? ????. ?? ??? ??? ?? ??? 10,000?? ??? ? train/validation? 8,000?/2,000?? ???, test ??? 2,000? ??? ????. ??? ??/? 2???

???? ?? ?? ?? ?? ?? ??? ??, ??, ??, ?? ????. ?? ?? ??? KoNLPy? ??? ?? ?? Okt ??? ?? ?? ?? ??? ????. ?? Keras TextVectorization?? ?? vocabulary 3,000?, sequence length 30? ?? sequence? ???.

2. ?? ??

??? ??

MLP baseline? 28 x 28 ??? 784?? ?? ?? ? Dense layer? ????. ??? Dense(256) - Dropout(0.2) - Dense(128) - Dense(10, softmax)??

CNN baseline? ??? 2?? ?? ?? 28 x 28 x 1 ??? ????. ??? Conv2D(32) - MaxPooling2D - Conv2D(64) - MaxPooling2D - Flatten - Dense(128) - Dropout(0.3) - Dense(10, softmax)??

??? ??

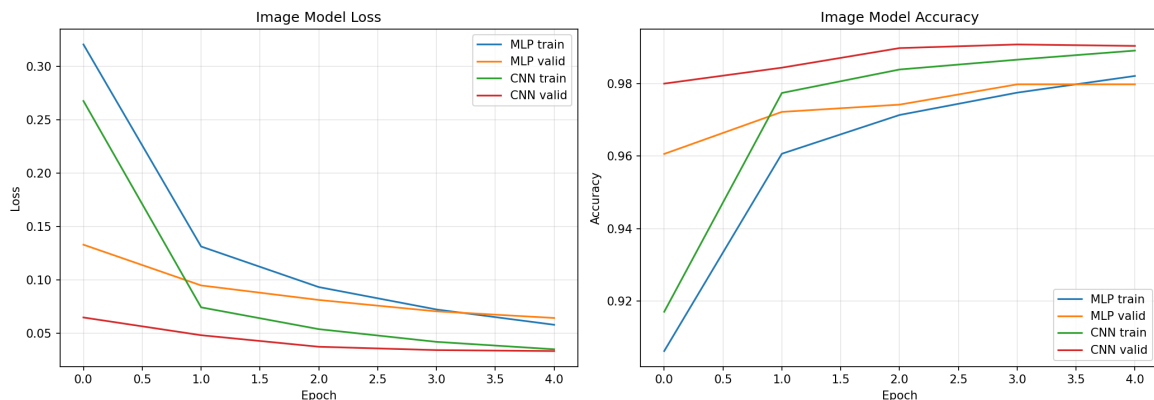
LSTM baseline? Embedding(64) ?? Bidirectional(LSTM(32))? ???, Dropout(0.3) - Dense(16) - Dense(1, sigmoid)? ?? ???.

Text CNN? Embedding(64) ?? Conv1D(64, kernel_size=3)? GlobalMaxPooling1D? ???, Dropout(0.3) - Dense(16) - Dense(1, sigmoid)? ?? ???.

3. ?? ??

??? ?? ??

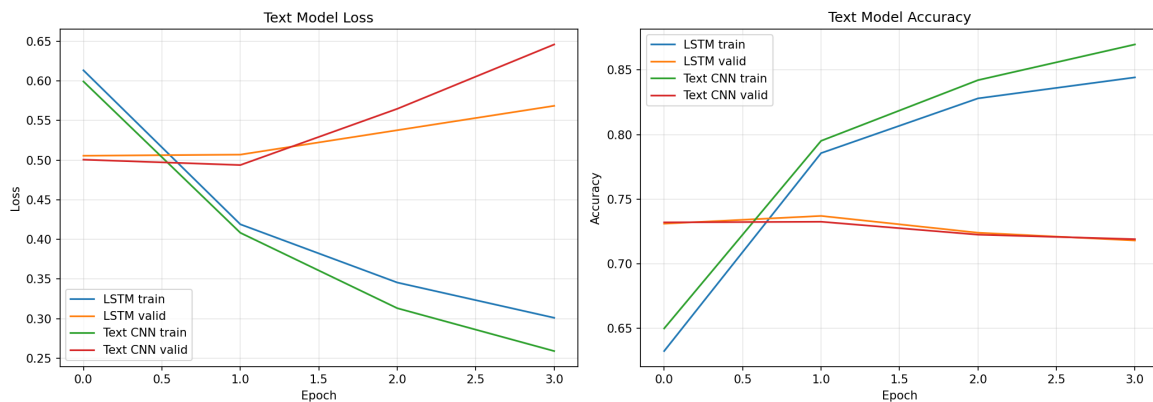
??	Train Accuracy	Validation Accuracy	Test Accuracy
MLP	0.9915	0.9798	0.9797
CNN	0.9945	0.9904	0.9899



CNN? MLP?? validation/test accuracy? ??? ? ?? ?? loss? ??? ????, CNN? test accuracy? ? 0.9899? MLP? ? 0.9797?? ???.

???

??	Validation Accuracy	Test Accuracy
LSTM	0.7370	0.7315
Text CNN	0.7325	0.7185



??? ????? train loss? ???. ?? validation loss? ?? ?? ??? ??? ??, 10,000? ?? ????? ??? ??? ?????. LSTM? test accuracy ? 0.7315, Text CNN? ? 0.7185? ???.

4. ??

????? CNN? ???? . MLP? ???? 1?? ??? ???? ??? ???? ?? ??? ?? ???? ???? . ?? CNN? ?? convolution filter? ??? ??? ????
?, ???, ?? ?? ?? ?? ???? ???? . MNIST ??? ?? ?? ??? ???? ???? ???? CNN? MLP?? ?? ??? ??? .

????? sequence ??? ?????. ?? ????? ??? ????? ??? ??? ??? ??? ??? ??? ??? ???. LSTM? ??? ?????? ??? ?? ?? ??? hidden state? ??? ? ???. Text CNN? ??? ??? ?????? ??, kernel_size=3? ?? ?? ??? n-gram ??? ?????.

? task? ???? ?? ?? ??? ?? ??? ???? ??? ??? ?? ??? ? ?? .???? 2?? ?? ??? ????? convolution? pooling? ?,?, ??? ???/?
 ??? ????? embedding ?? LSTM ?? sequence ???? Conv1D ?? Text CNN? ?.?. ??? ?? ?? ??? ???? ?? ??? ??? ?????
 ???? ??.